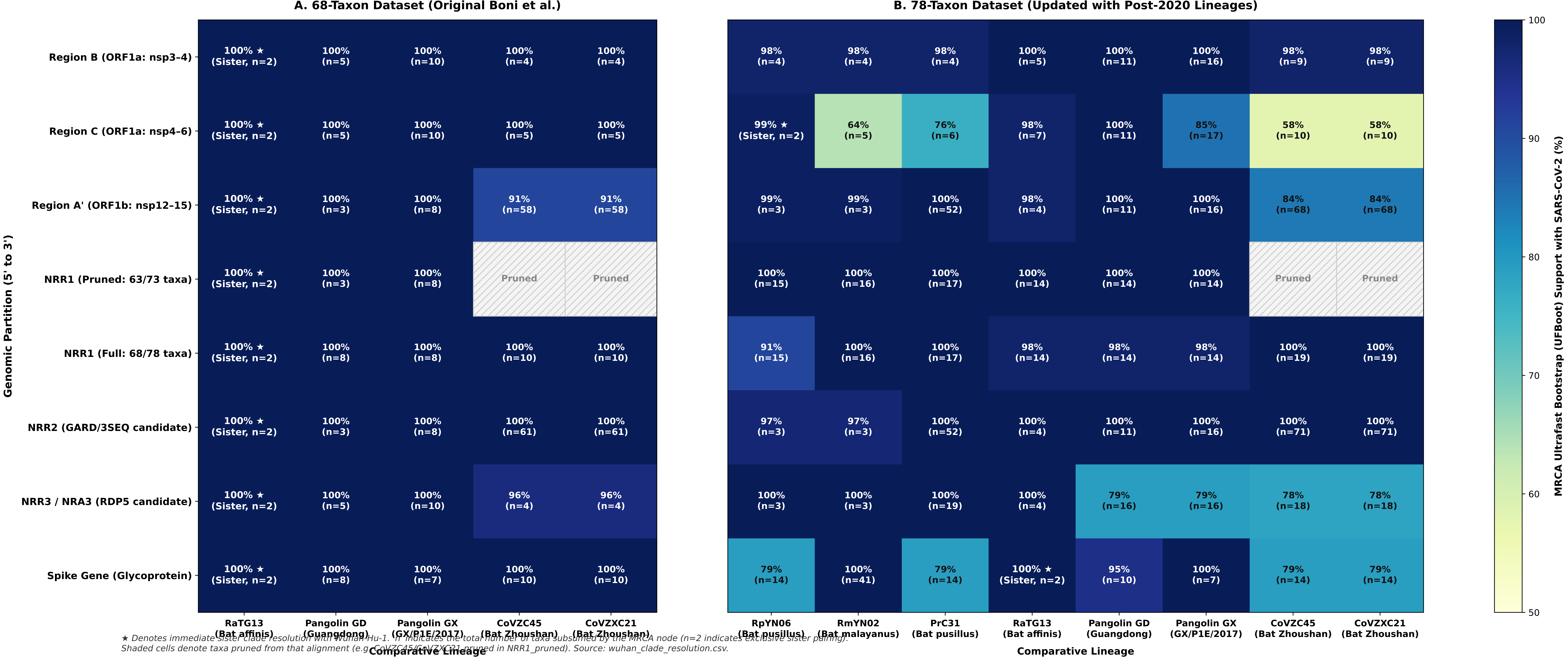
Comparative Differential Analysis of SARS-CoV-2 Clade Support and Node Topology Across Genomic Partitions in the Original (68-Taxon) vs. Updated (78-Taxon) Datasets



A. 68-Taxon Dataset (Original Boni et al.)

Genomic Partition (5' to 3')	RaTG13 (Bat affinis)	Pangolin GD (Guangdong)	Pangolin GX (GX/P1E/2017)	CoVZC45 (Bat Zhoushan)	CoVZXC21 (Bat Zhoushan)
Region B (ORF1a: nsp3–4)	100% ★ (Sister, n=2)	100% (n=5)	100% (n=10)	100% (n=4)	100% (n=4)
Region C (ORF1a: nsp4–6)	100% ★ (Sister, n=2)	100% (n=5)	100% (n=10)	100% (n=5)	100% (n=5)
Region A' (ORF1b: nsp12–15)	100% ★ (Sister, n=2)	100% (n=3)	100% (n=8)	91% (n=58)	91% (n=58)
NRR1 (Pruned: 63/73 taxa)	100% ★ (Sister, n=2)	100% (n=3)	100% (n=8)	Pruned	Pruned
NRR1 (Full: 68/78 taxa)	100% ★ (Sister, n=2)	100% (n=8)	100% (n=8)	100% (n=10)	100% (n=10)
NRR2 (GARD/3SEQ candidate)	100% ★ (Sister, n=2)	100% (n=3)	100% (n=8)	100% (n=61)	100% (n=61)
NRR3 / NRA3 (RDP5 candidate)	100% ★ (Sister, n=2)	100% (n=5)	100% (n=10)	96% (n=4)	96% (n=4)
Spike Gene (Glycoprotein)	100% ★ (Sister, n=2)	100% (n=8)	100% (n=7)	100% (n=10)	100% (n=10)

B. 78-Taxon Dataset (Updated with Post-2020 Lineages)

Genomic Partition (5' to 3')	RpYN06 (Bat pusillus)	RmYN02 (Bat malayanus)	PrC31 (Bat pusillus)	RaTG13 (Bat affinis)	Pangolin GD (Guangdong)	Pangolin GX (GX/P1E/2017)	CoVZC45 (Bat Zhoushan)	CoVZXC21 (Bat Zhoushan)
Region B (ORF1a: nsp3–4)	98% (n=4)	98% (n=4)	98% (n=4)	100% (n=5)	100% (n=11)	100% (n=16)	98% (n=9)	98% (n=9)
Region C (ORF1a: nsp4–6)	99% ★ (Sister, n=2)	64% (n=5)	76% (n=6)	98% (n=7)	100% (n=11)	85% (n=17)	58% (n=10)	58% (n=10)
Region A' (ORF1b: nsp12–15)	99% (n=3)	99% (n=3)	100% (n=52)	98% (n=4)	100% (n=11)	100% (n=16)	84% (n=68)	84% (n=68)
NRR1 (Pruned: 63/73 taxa)	100% (n=15)	100% (n=16)	100% (n=17)	100% (n=14)	100% (n=14)	100% (n=14)	Pruned	Pruned
NRR1 (Full: 68/78 taxa)	91% (n=15)	100% (n=16)	100% (n=17)	98% (n=14)	98% (n=14)	98% (n=14)	100% (n=19)	100% (n=19)
NRR2 (GARD/3SEQ candidate)	97% (n=3)	97% (n=3)	100% (n=52)	100% (n=4)	100% (n=11)	100% (n=16)	100% (n=71)	100% (n=71)
NRR3 / NRA3 (RDP5 candidate)	100% (n=3)	100% (n=3)	100% (n=19)	100% (n=4)	79% (n=16)	79% (n=16)	78% (n=18)	78% (n=18)
Spike Gene (Glycoprotein)	79% (n=14)	100% (n=41)	79% (n=14)	100% ★ (Sister, n=2)	95% (n=10)	100% (n=7)	79% (n=14)	79% (n=14)

★ Denotes immediate sister clade resolution with Wuhan-Hu-1. "n" indicates the total number of taxa subsumed by the MRCA node (n=2 indicates exclusive sister pairing). Shaded cells denote taxa pruned from that alignment (e.g. CoVZC45/CoVZXC21 pruned in NRR1_pruned). Source: wuhan_clade_resolution.csv.